

EMCH 260

SOLID MECHANICS

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Torsion

5

Chapter Objectives

- ✓ Torsion formula and shear stress distribution
- ✓ Polar moment inertia and application for circular cross section
- ✓ Angle of twist in single and multiple torques, and with different cross-sectional area
- ✓ Statically indeterminate torque-loaded members
- ✓ Shear flow and average shear stress in thin-walled tubes

TORSION FORMULA

- For linear elastic material, by applying *Hooke's law*, $\tau = G\gamma$, or $\tau_{\max} = G\gamma_{\max}$, we have the shear stress distribution

$$\tau = \left(\frac{\rho}{c}\right) \tau_{\max}$$

NOTE: Shear stress is in the plane perpendicular to the axis of the shaft

- For any cross section that is perpendicular to the axis of the shaft, the torque in the entire cross section is

$$T \equiv \int_A \rho(\tau dA) = \int_A \rho \left(\frac{\rho}{c}\right) \tau_{\max} dA$$

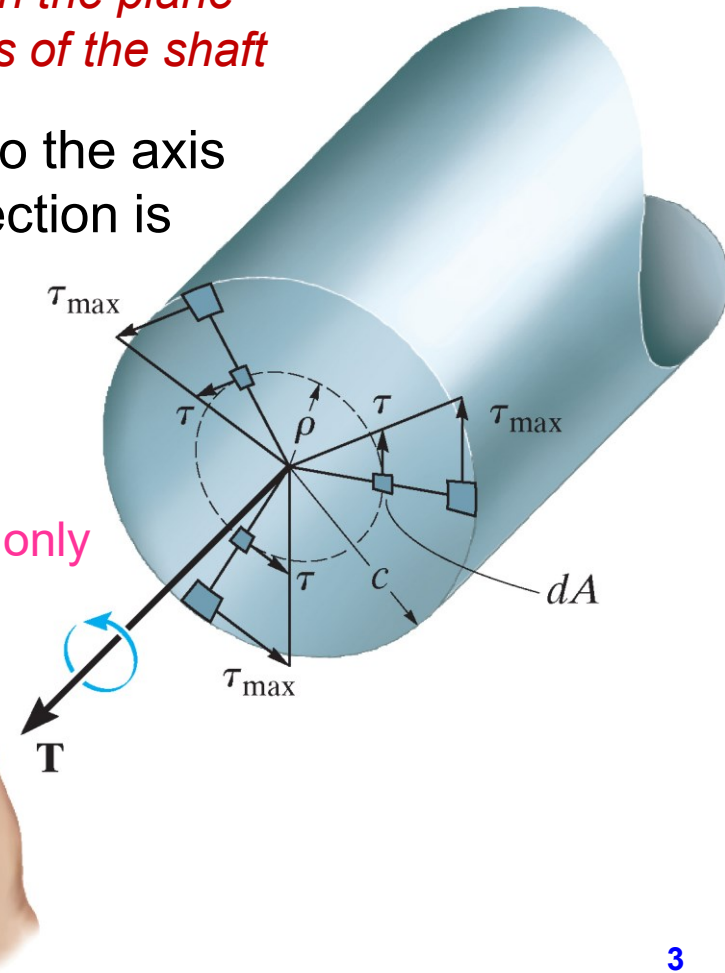
Internal resultant torque For circular cross-section only

$$\rightarrow \tau_{\max} = \frac{Tc}{J}$$

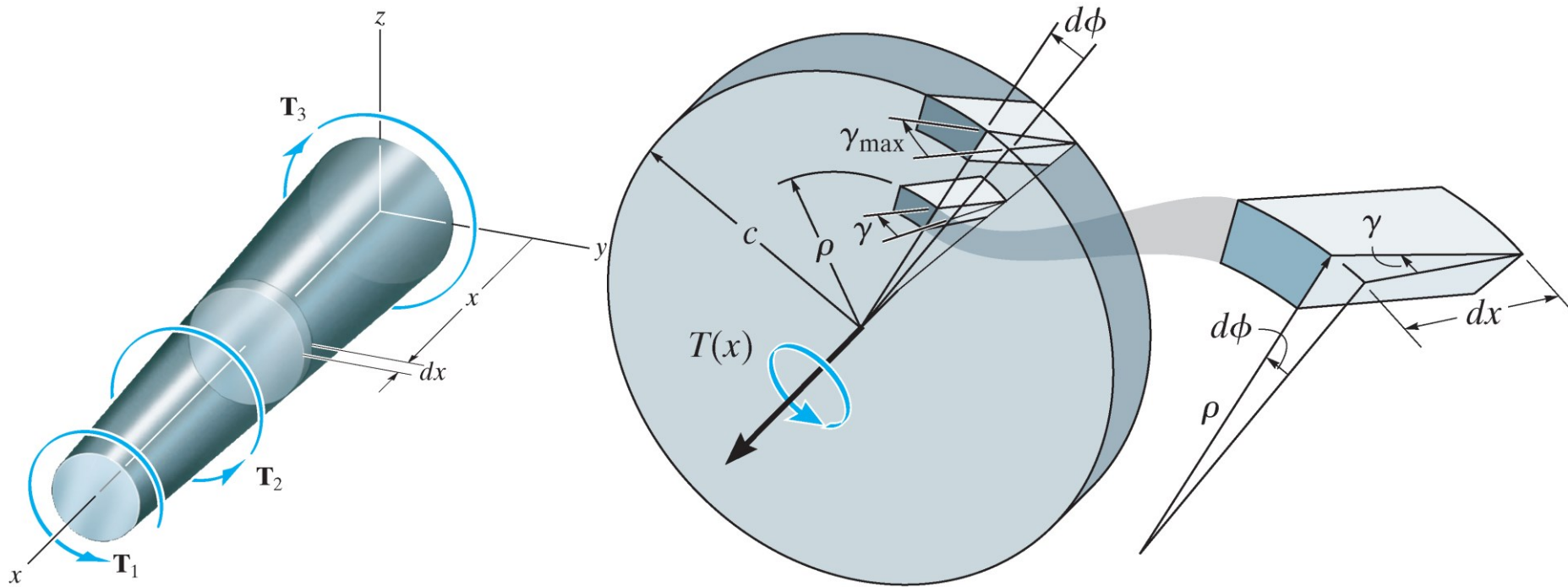
or

$$\tau = \frac{T\rho}{J}$$

J: polar moment of inertia



ANGLE OF TWIST



Integral form (twist angle of one end with respect to the other end):

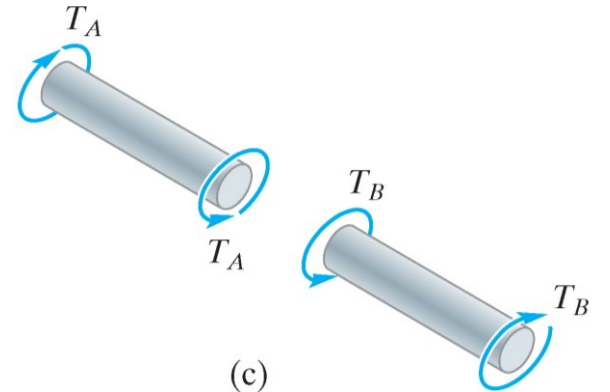
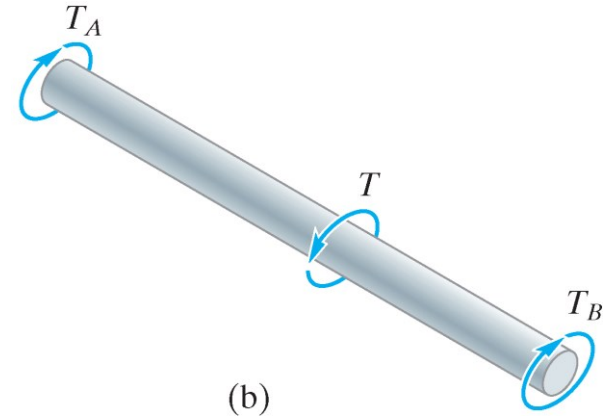
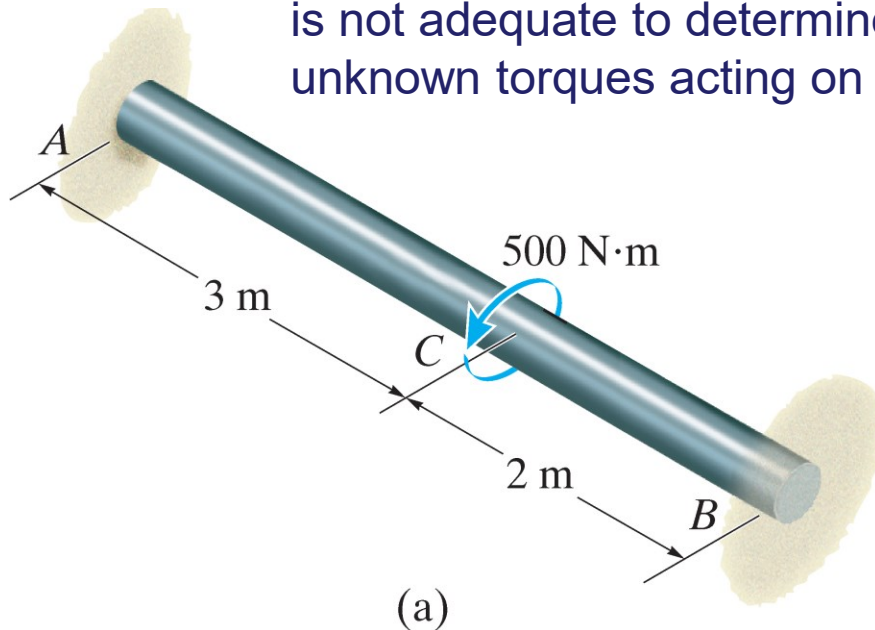
$$\phi \equiv \int_0^L \frac{T(x)}{J(x)G(x)} dx$$

$$\phi = \frac{TL}{JG}$$

Constant torque and cross-sectional area

STATICALLY INDETERMINATE TORQUE-LOADED MEMBER

- The moment equation of equilibrium is not adequate to determine the unknown torques acting on the shaft



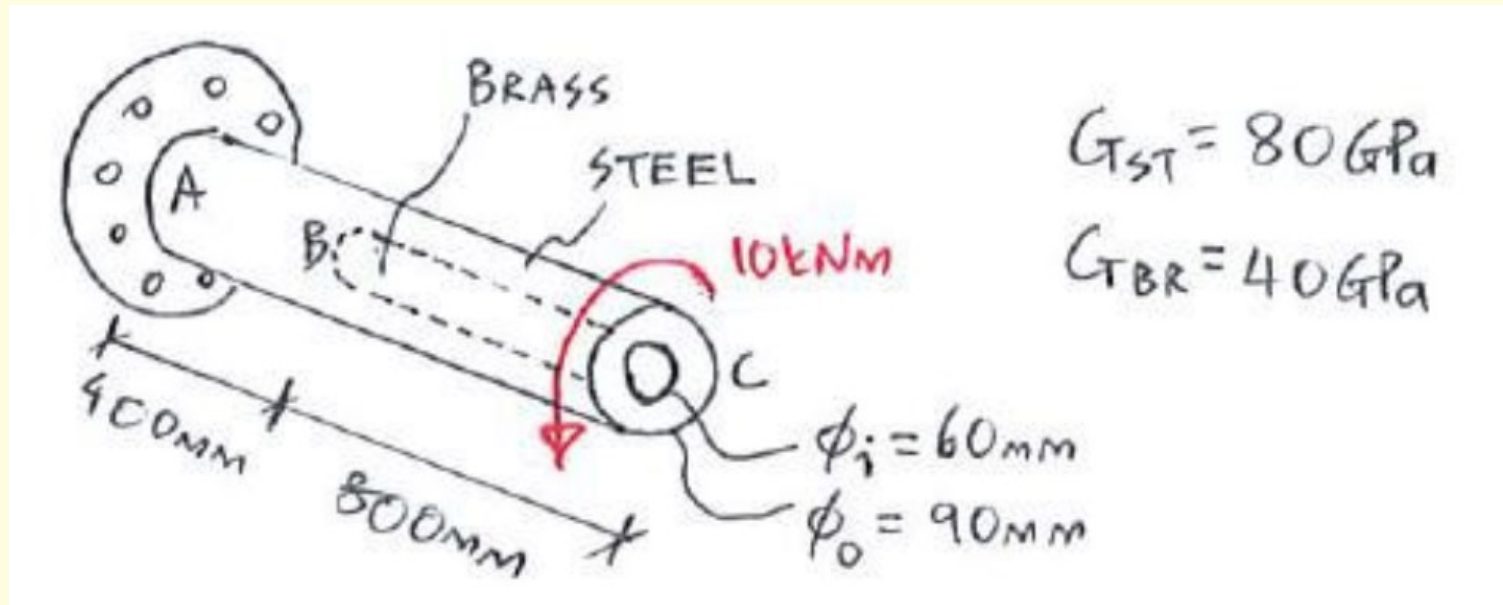
Compatibility Condition:

- Always think about twist angles!

$$\phi_{A/B} = 0$$

IN-CLASS EXAMPLE

The partial (BC) region of the shaft has a brass core with a diameter of 60 mm, with the outer region made from steel (outer diameter: 90 mm). The rest (AB) region of the shaft is solid and completely made from steel. The length of AB and BC region is 400 mm and 800 mm, respectively. The external torque at free end C is $10 \text{ kN} \cdot \text{m}$. The shear modulus of elasticity of steel and brass is 80 GPa and 40 GPa, respectively. The end A is fixed. Determine the angle of twist at end C relative to A .



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 1: Section Properties

Region AB (steel only): $J_{AB} = \frac{\pi}{2} c_1^4 = 6.4412 \times 10^{-6} \text{ m}^4$

Region BC (steel and brass):

$$J_{br} = \frac{\pi}{2} c_2^4 = \frac{\pi}{2} (0.06/2)^4 = 1.2723 \times 10^{-6} \text{ m}^4$$

$$J_{st} = \frac{\pi}{2} (c_1^4 - c_2^4) = 5.1689 \times 10^{-6} \text{ m}^4$$

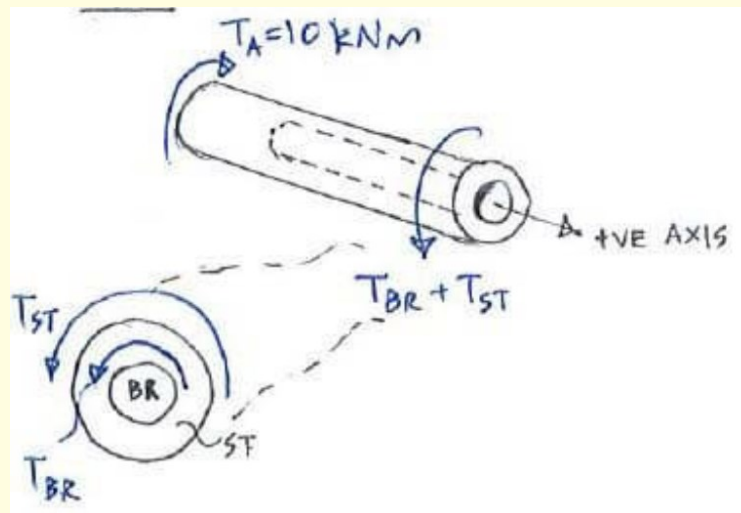
Step 2: Free-Body Diagrams

At end A or for region AB :

$$T_A = T_{AB} = 10 \text{ kN} \cdot \text{m}$$

For region BC :

$$\sum T = 0 \quad \rightarrow \quad T_{br} + T_{st} = 10 \text{ kN} \cdot \text{m} \quad (1)$$



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 3: Compatibility Condition

The twist angle of brass and steel in the region BC should be the same, i.e.

$$\phi_{br} = \phi_{st} \quad (2)$$

Step 4: Load-Displacement

Use load-displacement relations (such as $\phi = TL/(JG)$) to relate unknown twist angles to reactions.

$$\phi_{br} = \frac{T_{br}L_{BC}}{J_{br}G_{br}} \quad \phi_{st} = \frac{T_{st}L_{BC}}{J_{st}G_{st}}$$

Substituting into Eq. (2) and considering section properties, we have

$$\frac{T_{br}L_{BC}}{J_{br}G_{br}} = \frac{T_{st}L_{BC}}{J_{st}G_{st}} \quad \rightarrow \quad \frac{T_{br}}{1.2723 \times 10^{-6} \times 40 \times 10^9} = \frac{T_{st}}{5.1689 \times 10^{-6} \times 80 \times 10^9}$$

$$\rightarrow \quad T_{br} = 0.1231 T_{st} \quad (3)$$

Substituting Eq. (3) into Eq. (1), we get

IN-CLASS EXAMPLE (cont'd)

Solutions

Substituting Eq. (3) into Eq. (1), we get

$$T_{st} = 8904 \text{ N} \cdot \text{m}$$

$$T_{br} = 1096 \text{ N} \cdot \text{m}$$

Finally, the angle of twist at end C with respect to A is resulted from the torques in both region AB and BC .

$$\begin{aligned}\phi_{C/A} &= \phi_{AB} + \phi_{BC} = \frac{T_{AB}L_{AB}}{J_{AB}G_{st}} + \frac{T_{st}L_{BC}}{J_{st}G_{st}} \\ &= \frac{10000 \times 0.4}{6.4412 \times 10^{-6} \times 80 \times 10^9} + \frac{8904 \times 0.8}{5.1689 \times 10^{-6} \times 80 \times 10^9} \\ &= 0.025 \text{ rad} \quad \textbf{(Ans.)}\end{aligned}$$

FURTHER THINKING:

What if I changed the assumed direction of the resultant torque in brass? Will you get the same answers? Try to do it by yourself.